

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ

Chapter 22: Heights and Distances

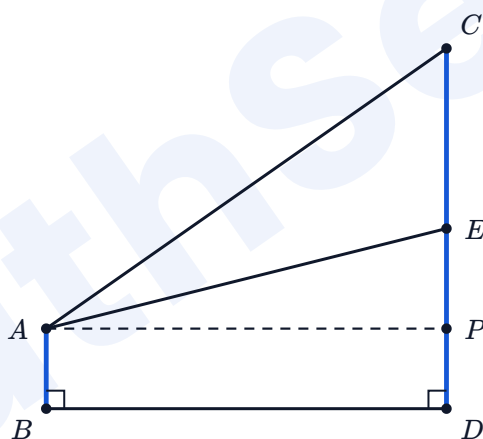
Previous year board questions, topic-wise: 4 questions

The tag after each question gives the board examination year and the marks. Solutions are in a separate PDF with the same question numbers.

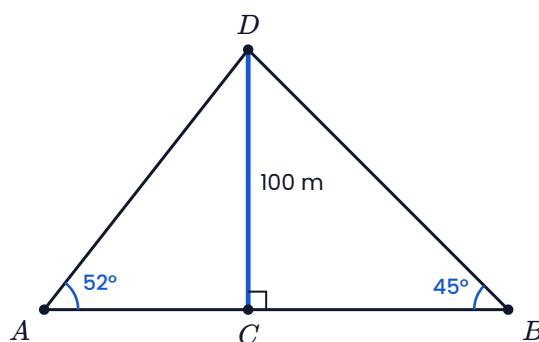
1 Problems with two triangles

1. A vertical tower AB stands 100 m from the foot D of a building CD . E is the mid-point of CD . The angle of depression of the top A of the tower is 35° from C and 14° from E . Using tangent values rounded to two decimal places, find the height of (a) the tower (b) the building.

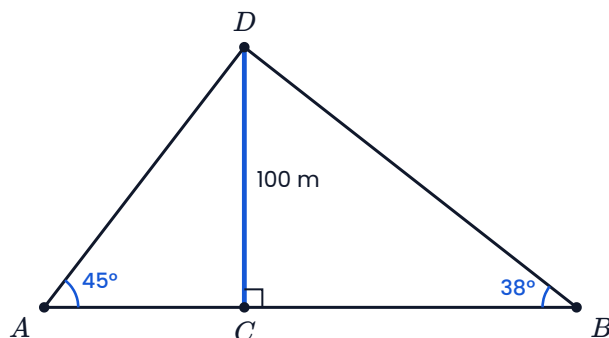
ICSE 2025, 5 marks



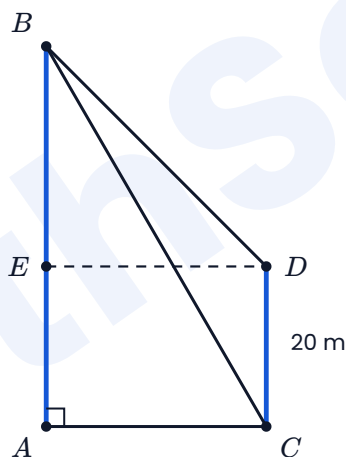
2. A tree CD is 100 m high. From two points A and B on the ground on opposite sides of the tree, its top D is seen at angles of elevation of 52° and 45° . Find the distance AB to the nearest metre. ICSE 2024, 5 marks



3. From the top D of a tower CD 100 m high, two ships A and B on opposite sides of the tower are seen at angles of depression of 45° and 38° . The ships and the foot of the tower are in one straight line. Find the distance between the ships to the nearest metre. **ICSE 2023, 4 marks**



4. From the top of a cliff, the top and the bottom of a tower 20 m high are seen at angles of depression of 45° and 60° . Find (a) the height of the cliff (b) the distance between the cliff and the tower. **ICSE 2020, 4 marks**



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